Josh Denbow | CS499 | Milestone 3 Narrative | 10/3/2026

The artifact I selected is, once again, the Android inventory management application originally created in CS 360: Mobile Architect and Programming. The application was originally designed to let users add, update, delete, and track inventory items. I continued using this artifact for my capstone because it provides several opportunities to improve an existing program while demonstrating different computer science skills.

In essence, I selected this artifact for the algorithms and data structures category because inventory management provides a more practical use for prioritization. For this specific enhancement, I added a reorder threshold to each inventory item and created a restock recommendation system. An item qualifies for restocking when its quantity is at or below its threshold, and its shortage is calculated by subtracting the current quantity from the reorder threshold.

The main algorithm uses a Java *PriorityQueue* and a comparator to rank qualifying items by their shortage, with the largest shortage receiving the highest priority. I also added tie-breaking rules, restock-priority sorting, threshold-based filtering, and a dedicated Restock Recommendations section in the interface. The recommendation logic remains in the service layer instead of the user interface, which keeps the program more organized and reusable.

This enhancement met the outcomes I planned to address in Module One. It especially supports the outcome of designing and evaluating computing solutions using algorithmic principles because I selected a *PriorityQueue* for the ranking problem and considered its efficiency and behavior. It also supports the outcome involving computing techniques and tools because the algorithm was integrated into a working Android application rather than being implemented by itself. My overall outcome coverage plan has not needed any major changes since I originally created it in Module 1.

This enhancement helped me see how a data structure can be used to solve a practical problem instead of only being demonstrated in an isolated example. One challenge was replacing the previous hard-coded low-stock rule with item-specific thresholds throughout the model, database, service, and interface. Testing different quantities and thresholds also helped confirm that the queue was ranking items based on actual shortage rather than simply sorting by quantity. Overall, the enhancement made the inventory app more useful while giving me a clearer example of algorithm and data structure selection.